

Artical Writing ▾

○ Methodology

Date limite

17 déc., 23:59 ↺ ▾

☰ Description

Visualisations : Not done yet ❌

1. Overview of the DCFL framework :
2. Testing process of every data-centric step
3. Model architecture enhancement
4. class distributions
5. Time series analysis of key features and failures
6. correlations and model evaluations plots...

Notes :

- add a Feature validation part in the feature selection section : using PCA and Correlation analysis in an irritative approach.\
- the intelligent downsampling should be applied only for the training set, the test set should be untouched.
- i should investigate the choice and testing of every chosen step.
- give result of feature selection benchmarking
- investigate how should i apply data balancing
- for the evaluation of the proposed model accuracy, we can compare the result of three stages. compare the performance of the centralized local model VS per-client local model VS global model. (the centralized model will be the maximum accuracy that can be reached via the model.)
- we will compare different out-of-failure periods for the normal class in the downsampling process to chose the right one.
- we can use TimeGAN to generate data of the minority class.
- to understand the failure behavior, we identified WT with two or more failures in the same component. and performed T-S and Trends Analysis. understanding the pre- and post-failure windows.

Strategies Evaluation :

1. Data labeling :

Evaluation no : SCADA concatenated + merged + labeled + handling missing values, of 30/60 pre-failure window + Evaluation

Combining Generator with G Bearing : Not done yet ❌

Model training revealed that the model always struggle to predicte the 4 class of the generator bearing cause it has only one failure in the entire 2year periode, and because of it happend in the very begining of the observed periode the validation and the test sets can't has samples of this failure class so we can't evaluate the model prediction ability on this class. that why we chosed to combain it with the generator failure class by the fact that they reference to almost the same component. with that we are overcaming the issue and the model can optimize more the prediction ability.

2. Post-Failure determination :

Data used :

SCADA concatenated + merged + labeled + Post-failure Injection+ handling missing values, of 30 or 60 pre-failure window + Evaluation

3. Feature selection :

use data :

data concatenated + merged + labeled + feature selection + handling missing values

4. Data Downsampling :

TimeGAN / Downsampling

5. Train Test Split :

The final two months could be removed, because they have only normal operation and won't be efficient for the test set.

- We have removed the last 2m, we used the remaining 3m for test and the 2m before as validation.
- the problem is that the train set has some small class failures that are located in the validation and test sets, which the model will not benefit from in the training.
- We should combine the generator and the generator bearing together to overcome the rare samples and the chances of predicting the generator bearing failures.

6. Model and hyperparameter selection : Not done yet ❌

- Model benchmarking and evaluation with the state-of-the-art models (we can go deeper by making it for every data-centric step)
- Hyperparameter tuning and tables presenting the resulting best parameters.
- final evaluation of the model based on another SCADA dataset.

7. Federated learning Evaluation : Not done yet ❌

- Test and evaluate different aggregation strategies (FedAvg, FedProx,...)
- The test set for both the federated models and the centralized model should be the same, eather using T11's data for both, or the same test set for both.

☑ Checklist

78%

- ☒ ~~add Class distribution of it~~
- ☒ ~~Label the entire 16/17 dataset~~
- ☒ ~~create two version of the labeled scada dataset using two pre failure windows 30 and 60days~~
- ☒ ~~created two version for each labeled scada dataset (30 and 60) of downsampled data using two periodes 6 and 8 month pre failure.~~
- ☒ ~~investigate different out of failure periodes~~
- ☒ ~~the baseline : data concatinated + merged + labeled + handling missing values~~
- ☒ ~~Present and showcase the Train Test Split, and explain how we'll use the Test set.~~
- ☐ We should combine the generator and the generator bearing together to overcome the rare samples and the chances of predicting the generator bearing failures.
- ☒ ~~Deal with missing value produced during the merging process~~

- ☒ Investigate the need to retain post-failure data points, and what range should i go with (if yes, redid the labeling process)
- ☒ Train and evaluate the model on every version of the labeled dataset, feature-selection and the downsampling period
- ☐ how to use TimeGAN in this case
- ☐ Start the Result Section
- ☐ Humanize and AI check the Content of the methodology section
- ☒ Redid data balancing section
- ☒ Redid the data labeling section
- ☒ refine the FL workflow subsection
- ☒ **Finishing-Methodology Todo List**
- ☒ Add Privacy methods and Non-IID data handling techniques

33%

- ☒ List of the chosen Pre-failure windows for the Dynamic Labeling process
- ☐ Do all the experiments and retrieve results (labeling, post-failure, Feature selection, data balancing, model selection)
- ☐ Add a choosen-approach section for every DC step
- ☒ From what exact periode we extract the validation set from the test set
- ☐ Hyper parramter turnning process, and the choosen parameters
- ☐ Do all the figures

☒ Labeling Evaluation

50%

- ☒ Train Model on Raw SCADA : Labeled 60/missing values handling
- ☒ Train Model on Raw SCADA : Labeled 30 /missing values handling
- ☐ Train Model on Raw SCADA : Labeled 60/Post-failure injection / missing values handling
- ☐ Train Model on Raw SCADA : Labeled 60/Post-failure injection / missing values handling / Combined Class 3 and 4 -> 3

☒ Feature Selection

25%

- ☒ Train Model on Knowledge-Based Feature Selection

- ☐ Train Model on Statistical Based Feature selection
- ☐ Train Model on Hybred (mixe strategy) Feature Selection
- ☐ Compare with the baseline
- ☒ **Data Balancing**

0%

- ☐ Train Model on Downsampling
- ☐ Train the model on balance data with TimeGAN
- ☐ Compare with the baseline

Commentaires et activité

 **yassire ammouri** a défini l'échéance de cette carte au mercredi à 23:59, avec la récurrence suivante : daily
[il y a 10 minutes](#)

 **yassire ammouri** a ajouté cette carte à Artical Writing
[13 oct. 2025, 18:07](#)